

SKILLS

Languages	Java, C, C++, C#, Python, JavaScript, SystemVerilog, RISC-V Assembly, Arm64
Software	Quartus, Questa, AutoCAD, Git, Office Suite
Electrical	Circuit Design and Components, Basic Circuit Tools, Oscilloscope

EDUCATION

Bachelor of Applied Science in Computer Engineering, <i>University of British Columbia</i>	2025 — Present
Bachelor of Science in Chemistry for the Medical Sciences, <i>University of Victoria</i>	2014 — 2018
Engineering First-Year Certificate, <i>Kwantlen Polytechnic University</i>	2024 — 2025
CGPA: 94%	

PROJECTS

Phonocardiogram Heart Sound Classifier — Python, scikit-learn, librosa, Claude Code 2026

- Built an end-to-end ML pipeline to classify heart sound recordings as Normal/Abnormal from the PhysioNet/CinC 2016 Challenge dataset (3,240 recordings), using Claude Code to accelerate implementation while directing architecture and validation decisions
- Diagnosed and fixed a data leakage bug where peak-detection-based segmentation introduced label-correlated artifacts, replacing it with fixed-length windowing validated against the full dataset
- Trained an SVM classifier on MFCC features achieving 0.87 balanced accuracy at the patient level, using patient-grouped cross-validation to prevent data leakage between train/test splits

Memory Chatbot — Python, Ollama, RAG 2026

- Built a locally-run AI chatbot in Python using Ollama with Llama 3, featuring persistent conversation memory saved to JSON and recalled by date
- Implemented Retrieval-Augmented Generation (RAG) to semantically search across past conversations and inject relevant context into new queries
- Designed a command-line interface supporting commands for conversation recall, semantic search, and user profiling via on-demand summarization

IoT Network Watchdog — C++, ESP32, JavaScript 2026

- Engineered a standalone IoT network monitor using an ESP32-WROOM and C++ to track connectivity status and latency, achieving high-frequency diagnostics with minimal system overhead
- Implemented persistent data logging using the LittleFS filesystem and NTP synchronization, enabling the device to maintain accurate, timestamped outage records in non-volatile flash memory across power cycles
- Designed an asynchronous web dashboard using JavaScript (ES6 Fetch API) and HTML5 to visualize live network metrics, ensuring real-time UI updates without interrupting background diagnostic tasks

UBC Course Planner — C#, .NET MAUI 2025

- Developed a responsive GUI for course planning application using .NET MAUI and implementing the MVVM architecture
- Implemented schedule saving and course conflict prevention

8-Bit Simple CPU — RISC-V, Altera FPGA 2025

- Built a custom 8-bit CPU from scratch using SystemVerilog featuring a dedicated Arithmetic Logic Unit (ALU) and dual-ported register file
- Constructed an ALU capable of performing addition, subtraction, multiplication, and bitwise AND operations, with integrated logic for condition flags (Negative, Zero, Carry, Borrow, and Overflow)
- Developed a 128-location memory system and implemented memory-mapped I/O to allow the CPU to interface with output registers and HEX displays

Personal Website — HTML, CSS alexng42.github.io 2025

- Developed and deployed a professional responsive portfolio website using HTML and CSS to showcase my software and hardware engineering projects
- Optimized user experience by applying responsive design principles, ensuring cross-browser and mobile compatibility

Tron Game — C, RISC-V, Altera FPGA 2025

- Developed a real-time light-cycle Tron game for an Intel DE10-Lite FPGA in C with RISC-V
- Implemented low-level collision detection logic by reading pixel colour data directly from the VGA frame buffer to trigger round-over states
- Improved system responsiveness by refactoring game engine to utilize interrupt service routines for user input and timer events